

**Time:** 5 minutes**Outcome:** Use Short-Run Firm Choice, Profit/Loss & Shutdown.**Difficulty:** ●●○ Intermediate

Short-Run Firm Choice, Profit/Loss & Shutdown



THE CORE IDEA

Short-Run Firm Choice, Profit/Loss & Shutdown uses $MR = MC$, firm demand, ATC, and AVC to determine output, price, profit or loss, and short-run shutdown. The firm shuts down when price cannot cover variable cost. At the chosen output, price above ATC means profit, price below ATC means loss, and equality means zero economic profit.



HOW TO RECOGNIZE IT

- Use marginal revenue and rising marginal cost to choose the firm's output.
- Read price and average total cost at that output to determine profit or loss.
- Use average variable cost to determine whether the firm operates or shuts down in the short run.
- Keep output choice, profit measurement, and shutdown analysis as three distinct steps.



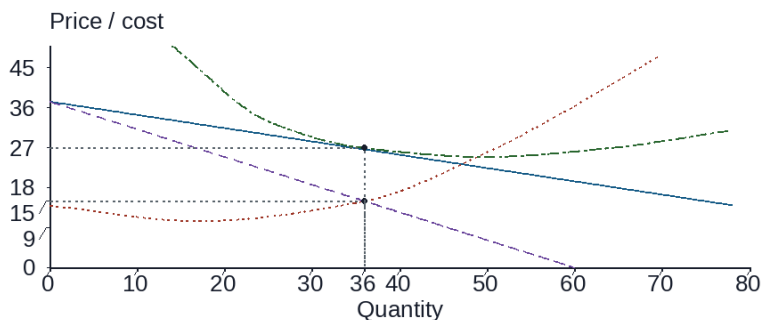
WATCH OUT

Separate profit from shutdown. A fixed-cost increase can create an economic loss without changing marginal cost or the benefit of operating; compare revenue with variable cost.



WORKED EXAMPLE: SHORT-RUN FIRM DECISION

— $D = AR$ - - MR ... MC - - ATC



The displayed baseline has $Q = 36$ and $P = ATC = \$27$, so economic profit is zero. Consider a higher fixed lease cost of \$108, with demand and variable costs unchanged. Output and price stay at that baseline, but ATC rises by \$3 and the loss is \$108. The firm still operates if revenue covers variable cost; loss alone does not imply shutdown.



CHECK YOURSELF

A firm earns a loss because P is below ATC . What additional cost comparison is needed to decide whether it should shut down in the short run?

**READY?**

Return to the game and master this concept.